



Microsoft Defender for Endpoint

Master Class

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Module 6

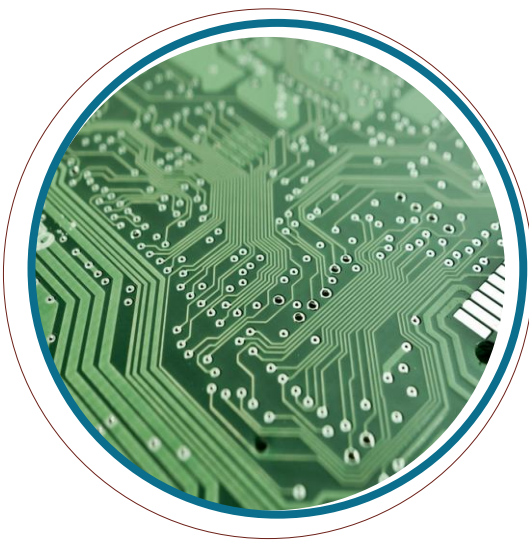
Advanced Hunting

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Module 5 Contents:

- **Advanced Hunting**
- **Kusto Query Language**
 - General
 - Data flow pipeline
 - Statements

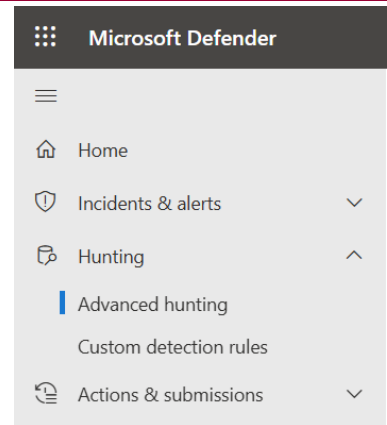


Advanced Hunting

Advanced Hunting

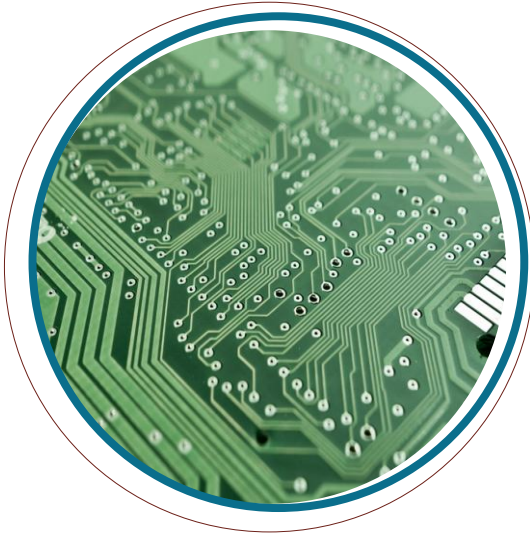
'... is a query-based threat hunting tool ...'

- Modes
 - guided
 - advanced
- Data freshness
 - Event and activity date (alerts, security events, ...)
 - immediately
 - Entity data
 - up to 24 hours



Advanced Hunting

- Schema - Tables
 - from multiple Defender Sources
 - Description in [documentation](#)
- Queries
 - could be saved query or function
 - added to schema for all administrators
- Result used
 - for investigation
 - Take actions
 - Detection rules



Kusto Query Language

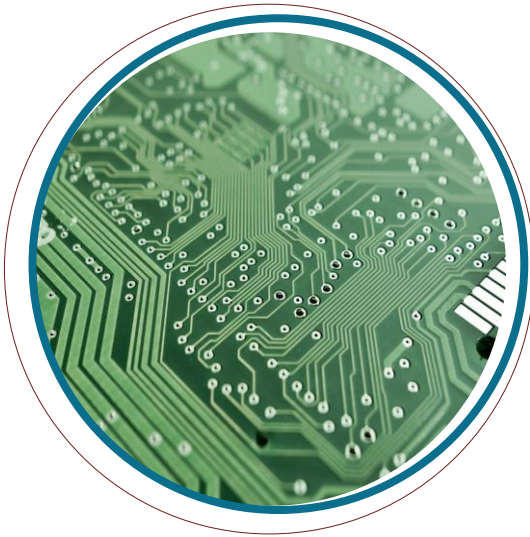
General

General

Some rules for KQL

- Case sensitivity
 - Tablenames
 - Fieldnames
 - Operators
 - Functions
- Comments: //
- Line breaks: before |
- Time: always saved as UTC

Source: <https://learn.microsoft.com/en-us/defender-endpoint/onboard-windows-client>



Data Flow pipeline

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Source: <https://learn.microsoft.com/en-us/defender-endpoint/overview-attack-surface-reduction>

Data Flow Pipeline

Overview

- Each statement starts with either a
 - Table
 - Variable(s) declaration(s)
 - Functions with result type table
- Variable declaration must end with ;
- Use | to pass data from table to operator
 - Multiple | are possible

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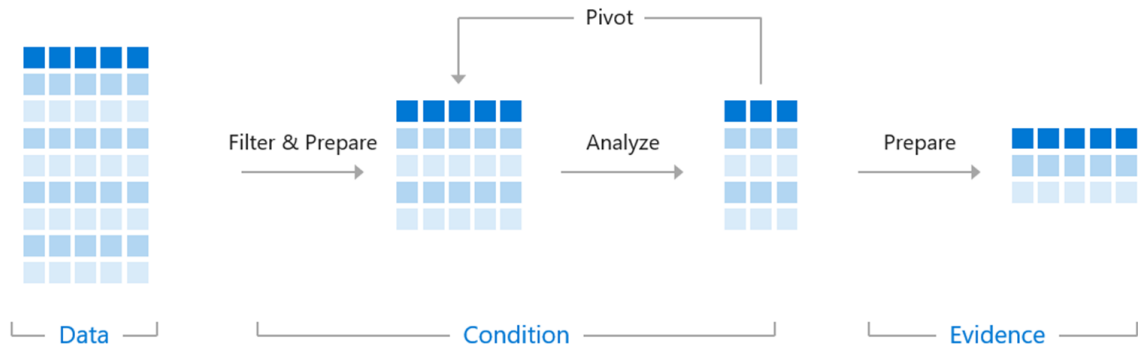
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Source: <https://learn.microsoft.com/en-us/defender-endpoint/overview-attack-surface-reduction>

Data Flow Pipeline

```
SecurityEvent | where EventID == "4626" | summarize count() by Account | limit 10
```

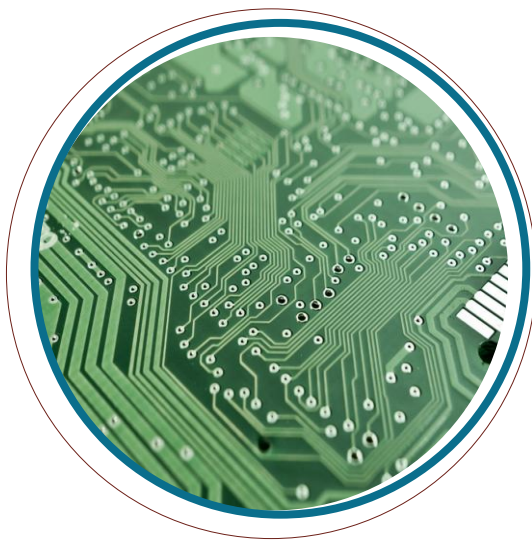


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Basic statements

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Basic statements

- Getting table data

```
DeviceInfo
// Case-sensitivity
```

- Using Pipeline

```
DeviceInfo
| limit 5
// limit doesn't sort the records.
// Alias for limit: take
```

```
DeviceInfo
| top 5 by DeviceName desc
// Records sorted first
```

Basic statements

- Sorting results

```
DeviceInfo
| sort by DeviceName asc
DeviceInfo
| sort by DeviceName asc, PublicIP desc
// Alias for sort: order
```

Basic statements

- Variables

```
let lmt = 3;
DeviceInfo
| sort by DeviceName asc , Timestamp desc
| limit lmt;
// let creates a variable.

let myTable =
DeviceInfo
| limit 10;
myTable
// A variable could also contain a table.
```

Basic statements

- design result

```
DeviceInfo
| project DeviceName,DeviceType,PublicIP
// only this columns appear in the result
```

```
DeviceInfo
| project-away DeviceType,PublicIP
// all columns of DeviceInfo except DeviceType,PublicIP appear in the result
```

```
DeviceInfo
| project-keep Device*,Device*,PublicIP
// project-keep has the same result as project but you could use *
```


Basic statements

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```
DeviceInfo
| project-keep Device*,Device*,PublicIP
// project-keep has the same result as project but you could use *
```

Basic statements

- Filter records

```
DeviceInfo
| where DeviceName =~ 'Client1'
// =~ case-insensitive
```

```
DeviceInfo
| where DeviceName startswith "server" // server*
```

```
DeviceInfo
| where DeviceName endswith ".local" // *.local
```

```
DeviceInfo
| where DeviceName contains "aztrg2112" // *aztrg2112*
```

Basic statements

- Extends result

```
AlertEvidence
| where EntityType == 'File'
| extend FileSizeKB = FileSize / 1024
| project FileName,FileSize,FileSizeKB
```

```
AlertEvidence
| where EntityType == 'File'
| extend FileSizeKB = FileSize / 1024,
|         FileSizeMB = FileSize / 1024 / 1024
| project FileName,FileSize,FileSizeKB,FileSizeMB
```

Basic statements

- Remove duplicates

```
DeviceInfo
| project DeviceName
| distinct DeviceName
```

```
DeviceInfo
| project DeviceName,PublicIP
| distinct DeviceName,PublicIP
```

```
DeviceInfo
| summarize by DeviceName,PublicIP
```

Basic statements

- Group records

```
DeviceInfo
| summarize count() by DeviceName
```

```
DeviceInfo
| summarize count() by DeviceName,PublicIP
```

```
DeviceInfo
| summarize Qty = count() by DeviceName,PublicIP
| sort by DeviceName asc, Qty desc
```

Basic statements

- Group records
- some aggregate functions

sum()	make_list()	arg_max()
avg()	make_set()	arg_min()
min()	make_bag()	
max()		

Basic statements

Join tables

- union
 - 'concatenates' two or more tables
- join
 - 'joins' two tables using key properties

Basic statements

- extracting text
 - extract() function - use regular expression
 - parse operator
- expanding arrays
 - mv-expand operator
- expanding json objects
 - parse_json() function to convert string to json object
 - evaluate operator + bag_unpack() function

Source: <https://learn.microsoft.com/en-gb/defender-xdr/advanced-hunting-schema-tables>

Device Tables

Devices		^
✓	DeviceEvents	⋮
✓	DeviceFileCertificateInfo	⋮
✓	DeviceFileEvents	⋮
✓	DeviceImageLoadEvents	⋮
✓	DeviceInfo	⋮
✓	DeviceLogonEvents	⋮
✓	DeviceNetworkEvents	⋮
✓	DeviceNetworkInfo	⋮
✓	DeviceProcessEvents	⋮
✓	DeviceRegistryEvents	⋮

Device Tables

DeviceInfo

Machine information, including OS information

DeviceId, DeviceName, PublicIP, OSPlatform, IsExcluded, DeviceType, JoinType, LoggedOnUsers, ...

DeviceEvents

Multiple event types, including events triggered by security controls such as Windows Defender Antivirus and exploit protection

ActionType, FileName, ProcessID, ProcessCommandLine, Registry*, ...

Device Tables

DeviceFileEvents

File creation, modification, and other file system events

DeviceId, DeviceName, ActionType, FileName, FolderPath, InitiatingAccount*, InitiatingProcess* ...

DeviceLogonEvents

Sign-ins and other authentication events

ActionType, DeviceName, LogonType, Account*, Remote*, AdditionalFields, ...

Device Tables

DeviceNetworkInfo

Network properties of machines, including adapters, IP and MAC addresses, as well as connected networks and domains

DeviceNetworkEvents

Network connection and related events

Remote*, Local*, ...

Device Tables

DeviceProcessEvents

Process creation and related events

DeviceRegistryEvents

Creation and modification of registry entries

DeviceImageLoadEvents

DLL loading events

RegistryKeyCreated,
RegistryKeyDeleted,
RegistryKeyRenamed,
RegistryValueDeleted,
RegistryValueSet



Some Examples

Try to read the query

- Start PowerShell without profile

```
DeviceProcessEvents
| where ProcessCommandLine has_all ("-nop", "powershell.exe")
| summarize TotalCommands = dcount(ProcessCommandLine),
|           ExecutedCommands = make_set(ProcessCommandLine) by DeviceName
```

- Disable Defender

```
DeviceProcessEvents
| where FileName =~ "powershell.exe"
| where ProcessCommandLine has_any ("Add-MpPreference", "Set-MpPreference")
| where ProcessCommandLine has_any ("ExclusionProcess", "ExclusionPath")
```

Try to read the query

- Last information of devices by name

```
DeviceInfo
| where isnotempty(DeviceName) and isnotempty(OSPlatform)
| summarize arg_max(Timestamp, *) by DeviceName
```

- Last network information of non-excluded devices by DeviceID

```
DeviceNetworkInfo
| join DeviceInfo on DeviceId
| where not(IsExcluded)
| summarize arg_max(Timestamp, *) by DeviceId
```


Try to read the query

- Webrequests by PS

```
DeviceEvents
| extend _tmp = parse_json(AdditionalFields)
| where _tmp.Command contains 'Invoke-WebRequest'
| order by Timestamp
```

- Open service ports

```
let RemoteServices = dynamic([22, 139, 445, 3389, 5900, 5985, 5986]);
DeviceNetworkEvents
| where ActionType == "ListeningConnectionCreated"
| where LocalPort in (RemoteServices)
| summarize OpenPorts = make_set(LocalPort),
              TotalOpenRemoteServicesPorts = dcount(LocalPort) by DeviceName
| sort by TotalOpenRemoteServicesPorts
```

More queries and hints for hunting

- <https://github.com/Bert-JanP/Hunting-Queries-Detection-Rules/tree/main/Defender%20For%20Endpoint>
- <https://github.com/Bert-JanP/Hunting-Queries-Detection-Rules>



End of Module 6